

6.4.2 Fundamental particles

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) particles and antiparticles; electron–positron, proton–antiproton, neutron–antineutron and neutrino–antineutrino	
(b) particle and its corresponding antiparticle have same mass; electron and positron have opposite charge; proton and antiproton have opposite charge	
(c) classification of hadrons; proton and neutron as examples of hadrons; all hadrons are subject to the strong nuclear force	
(d) classification of leptons; electron and neutrino as examples of leptons; all leptons are subject to the weak nuclear force	
(e) simple quark model of hadrons in terms of up (u), down (d) and strange (s) quarks and their respective anti-quarks	

- (f) quark model of the proton (uud) and the neutron (udd)
- (g) charges of the up (u), down (d), strange (s), anti-up ($\bar{u}$), anti-down ($\bar{d}$) and the anti-strange ($\bar{s}$) quarks as fractions of the elementary charge e
- (h) beta-minus (β^-) decay; beta-plus (β^+) decay
- (i) β^- decay in terms of a quark model;

$$d \rightarrow u + {}^0_{-1}e + \bar{\nu}$$
- (j) β^+ decay in terms of a quark model;

$$u \rightarrow d + {}^0_1e + \nu$$
- (k) balancing of quark transformation equations in terms of charge
- (l) decay of particles in terms of the quark model.

